

MAINS MASTER PROGRAM (MMP) 2024

ECONOMY-15; INFRASTRUCTURE-3: AIRWAYS AND ENERGY SECTOR

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1) AVIATION SECTOR IN INDIA

- **Introduction**
 - » India is the **third largest domestic aviation market** (After USA and China). It contributes to **\$35 billion** annually to India's GDP and offer 7 million jobs.
 - » **Steady growth in the sector**: The sector grew with a rate of 14% per annum between FY14 and FY20. The number of airports has doubled since 2014.
 - » **Improving Safety**: India has improved its ranking to 48 in ICAO's Global safety ranking.
- **The sector has a lot of potential to further grow in India:**
 - » India's growing middle-class population and improving socio-economic conditions including high disposable income.
 - » **Growing tourism sector**
 - » **Favorable demographics** -> Younger population prefer to fly more
 - » **Improved penetration of airways** due to initiatives like UDAN (Regional Connectivity Scheme).
- **Risks / Limitations/ Problems of India's Aviation Sector**
 - » **Infrastructure inadequacy**-> The current airports will not be able to take the load of increasing demand of future.
 - » **Absence of connectivity hubs in India** -> A large portion of Indian international traffic goes through connectivity hubs in the middle east and southeast Asia.
 - » **Human resource shortage** -> other than shortage of pilots and technical staff there is also an emerging shortage of skills.
 - » **High Jet fuel prices** -> increases the cost of operations of the airlines.
 - » **Constraint finances of the sector**: Due to intense competition, high accumulated losses.
 - » **Penetration of aviation sector** has improved due to UDAAN but is still **very limited** and covers very few regions of the world.
 - » **Flawed policy and regulatory framework**
- **Way Forward:**
 - » **Augmenting the Airport Capacity**: India's future demand of civil aviation is going to be high and current infrastructure is incapable of handling it.
 - » **Revise PPP norms** to attract more private players.
 - » **Focus on human resource development and skill development** in the civil aviation sector, with special emphasis of recently emerging drone sector.
 - » **Need of providing adequate long-haul connectivity** from India by strengthening Indian airlines.
 - » **Maintenance, Repair and Overhaul (MRO)** services need to strengthen to fuel the growth in the sector. For this human resource creation with appropriate skills will also be crucial.
 - » **Focus on efficiency and environmental sustainability** to ensure long term viability of the project.

2) NATIONAL CIVIL AVIATION POLICY, 2016 (NCAP)

- Introduction

- The NCAP, 2016 is the first civil aviation policy launched in India. It **aims** to promote regional connectivity; take flying to masses by making it affordable; establish an integrated ecosystem which will lead to significant growth in civil aviation sector and promote tourism, employment and balanced regional growth; enhance ease of doing business through deregulation, simplified procedures, use of technology etc.

- Salient Features

- **Regional Connectivity Scheme (UDAN Scheme)** - through subsidization of regional flights.
- **Route Dispersal Guidelines** - to make more services available for Tier-2, Tier-3 cities
- **5/20 Requirements replaced with new schemes for starting international flights -> Provides level playing field for new players**
 - » All airlines can commence international operations provided that they deploy 20 aircraft or 20% of total capacity (in terms of average number of seats on all departure put together), whichever is higher in domestic operations.
- **Bilateral Traffic Rights** - Enter into 'OPEN Sky' agreements with SAARC and other neighboring countries ASAP.
- **Airport development through PPP mode, by AAI, state governments or private sector**
 - » Future tariffs at all airports will be calculated on a '**hybrid till**' basis, unless specified otherwise in concession agreements.
 - 30% of non-aeronautical revenue will be used to cross subsidize aeronautical charges
- **Aviation Education and Skill building**
 - » All training in non-licensed category will conform to National Skill Qualification Framework standards.
 - » MoCA will provide full support to Aviation Sector Skill Council and other similar organizations/agencies for imparting skills for the growing aviation sector

3) REGIONAL CONNECTIVITY SCHEME (UDAN SCHEME)

- **Background: India's aviation sector suffered from the following issues before the launch of regional connectivity scheme (RCS):**
 - » **Traffic concentrated between large airports:** It contributed to choking of large airports
 - » **Existence of a lot of ghost airports:** Inefficient resource utilization
 - » **Poor connectivity to tier-2 and tier-3 cities** -> this also hampers other services and tourism; negatively impacts national integration.
 - » **States have ignored aviation sector** as Civil aviation is mentioned in the Union List.
- **UDAN Scheme** (A part of National Civil Aviation Policy 2016)
 - » Launched in 2017 with an aim to provide air connectivity to tier II and Tier-III cities and ensuring affordable fares.
 - » It focuses on revival of existing unserved/underserved airports/airstrips of the state governments, PSUs, AAls etc.
 - » Government also approved several green field airports across the country to improve rural connectivity.
- **Key Provisions of the scheme**
 1. **Subsidization of regional flights - to promote regional connectivity**
 - » **Regional Connectivity Fund (RCF)** has been created to meet the VGF requirement.
 2. **Cap on Airfare** at Rs 2,500/hour journey for 50% of the seats.
 3. **Other steps** which are being taken
 - » **Revival of airstrips/airports as No-Frills Airports** at an indicative cost of Rs 50 crore to Rs 100 crore
 - » No airport charges
 - » Reduced excise duty of 2% on ATF picked at RCS airports
 4. **Equitable Geographical Allocations:** For balanced regional growth, allocations will be spread equitably across 5 regions - North, South, West, East and Northeast with a cap of 25%.
 5. Regional connectivity scheme is in operation for **10 years** with individual route contract for a 3-year span.
 6. Market based **reverse competitive bidding mechanism** to determine least VGF to select the airline operator with the right to match the initial proposer.
- **Key Achievements/Significance of UDAN Scheme:**
 - » Increased number of operational airports to 140+ from 74 in 2014
 - » In last 7 years (2017-2024), since the commencement of UDAN, **1390 valid awarded routes** have been allotted to various airlines. Out of these 579 RCS routes connecting 85 unserved and underserved airports have been operationalized.
 - » Further, around 50 helicopter routes have also been made function (UDAN 5.1 was specifically focused on Helicopters)
 - » All this has led to **inclusive growth**, improved employment and more tourism.
- **Some Challenges**
 - » **Not all the underserved and unserved (ghost) airports** which were envisaged to become operational have been revived and **not all the routes** for which airlines (new and established ones) had placed bids have been started.
 - » **New players** (like Air Deccan and Air Odisha) have little experience, lack funds and are struggling with economics and logistics of providing connectivity from remote locations. **Quality of service**

has also remained a concern with highest number of cancellations being seen for these new players.

- » **Concerns about sustainability after VGF funding ends** in 3 years is also keeping the aviation player worried especially now when oil prices are seeing a positive trend.
- » **States have been reluctant to contribute**
 - For e.g. VAT on ATF has not been decreased by many.
- » **Land Acquisition for greenfield airports** have emerged as a major issue.

- **Some criticisms of the scheme**

- » Subsidy for air travel will only benefit middle class and upper middle class -> it could be used at other places
- » Before COVID-19 crisis, India had the fastest growing aviation market and thus state subsidies could be avoided.

- **Way forward**

- » Government should think about extending the VGF beyond three years to bring in more player at the route.
- » Regulations for Scheduled helicopter service should be finalized ASAP so that smaller airports can be viably connected.
- » There is a need of **higher investment on airport infrastructure** to increase the quality of service and to complement the efforts by new players in the aviation industry.
- » **A Cooperative Federalism Framework** can provide the required impetus to the Civil Aviation Sector.
 - How can states contribute:
 - Reduce VAT on ATF
 - Develop airports on its own or through PPP

- **Conclusion**

- » Travel dynamics through India is evolving and air service development is becoming a major force in this development. The newly connected cities have massive opportunities to boost regional and inter-continental connectivity.
- » And with the significant benefits that a new air service brings to community, UDAN will only help to facilitate the further evolution of a modern, well-connected country.

4) PPP IN AVIATION SECTOR

- **PPP Definition:**

- » Public-Private Partnership (PPP) models have been utilized in the development of airports in India to leverage private sector expertise, investment, and efficiency while ensuring public control and oversight.

- **Various PPP models used in airport development in India are:**

- i. **Build Operate Transfer (BOT):** E.g. Delhi International Airport Limited (DIAL): DIAL was formed as a JV between AAI and GMR Group.

- ii. **Design-Build-Finance-Operate-Transfer (DBFOT):** This model is similar to the BOT model, but the private entity is also responsible for the design and construction of the airport facility. E.g. The Bangalore International Airport Limited (BIAL).
 - iii. **Operate Maintain Transfer (OMT):**
 - E.g. Chhatrapati Shivaji International Airport (CSIA), Mumbai
 - It is owned by AAI, and Mumbai International Airport Limited (MIAL), a joint venture between GVK-led consortium and AAI, operates and manages the airport. MIAL receives the management fee based on performance targets.
 - iv. **Joint Venture:** E.g. Cochin International Airport Limited (CIAL) was formed by a joint venture between the government of Kerala and a consortium of NRIs. The consortium provided the necessary capital, and the airport was developed and operated jointly by the government and the consortium.
- **Critical analysis of development of airports in India using JV:**
- i. **Positives:** PPP models has helped in improving infrastructure and services at some of India's busiest airport, without increasing government's fiscal burden.
 - For e.g. airports in Delhi and Mumbai have undergone major expansion and renovations under PPP model.
 - Government is able to focus on regional connectivity more.
 - ii. **Challenges/Limitations:**
 - It has been difficult to attract private investors to airports as airports are often seen as risky investments, as they are subjected to fluctuation in air traffic.
 - **Higher airport charges for passengers**
 - **Other general challenges with PPP** (done in previous topics - Contract issues etc.)
- **Overall**, the development of airports using PPP model has been a mixed bag in India.

5) DRONE SECTOR

- **Example Questions**
 - » "Drones are frontier technology which has the potential to leapfrog India's economic growth" Elaborate [10 marks, 150 words]
 - » Discuss the key challenges limiting the growth of drone technology and its usage in India [10 marks, 150 words]
- **Background:** Remotely Piloted Aircrafts, popularly referred as drones, are unmanned aircrafts piloted from a remote platform. It has many wide-ranging applications. But, due to various security concerns, the **GoI had, in Oct 2014, banned the use of UAVs by private players and individuals**. After years of deliberations, the new regulations by DGCA which came into force on 1st Dec, 2018 have lifted this ban. It has led to revival of drone sector in the country.
- **Key Areas of Application:**
 - » **Agriculture:** Surveillance (plant counting, estimating crop yield, monitoring); Fertilizer, pesticide Supply.
 - **Kerala Agriculture University (KAU)** has recently used drones (drone seeder) to sow paddy to save time, money, and seeds.

- Government of India has recently launched a central sector scheme (NAMO Drone Didi) to provide drones to the Women Self Help Groups (WSHGs). This will be used by SHGs to provide on rental to farmers for agri-purposes.
- » **Media and Entertainment:** Aerial documentary, photography, videography
- » **Mining:** Thermal imaging, terrain mapping etc.
- » **Infrastructure Development:** Industrial inspections, 3-D video mapping, Land Audit, town planning etc.
- » **Energy and Utilities:** Site monitoring, survey grade maps for siting transmission lines.
- » **E-Commerce:** Transportation of deliveries
- » **Defence and Policing:** Replacing manned aircraft in future, Border security forces, coast guards, navies etc may use it for surveillance; general monitoring of law and order situation. E.g. The National Aerospace Laboratories (NAL) has successfully completed the first test of a solar powered drone which functions as "pseudo satellite" - The High Altitude Pseudo Satellite Vehicles (HAPS). This can significantly increase India's surveillance and monitoring capabilities.
- » **Health Sector:** supplying of medicines to remote areas.
- **Challenges and limitations hindering the growth of drone sector in India**
 - » **Evolving Regulatory framework:** The changing policies can hamper ease of doing business.
 - » **Safety Concerns:** More drone usage can lead to drone falling/accidents which may create problems specially in densely populated area.
 - » **Privacy Concerns:** Drones can be used to spy on people leading to violation of right to privacy.
 - » **Cyber security issues:** Since drones are part of connected system, hackers may end up controlling drones and thus use it for illegal purposes.
 - » **Job losses:** With the penetration of drones in agriculture, delivery services etc., the jobs availability in the country may get hampered.
 - » **Military use of drones can also be very devastating:**
 - **Easier, violation of sovereignty**
 - For e.g. in recent years, sightings of drones along the Indo-Pak borders have increased.
 - Drones have been used by Pakistan based outfits to smuggle arms, ammunitions and drugs in India.
 - **Risk of over using of military options**
 - Making it more difficult to identify violations of constraints against targeting non-combatants
- **Key steps taken in recent times to promote drone sector in India:**
 - » Liberalized Drone Rules, 2021
 - » Release of India's Airspace Map for Drone Operations.
 - » Implementation of PLI Scheme for drone sector.
 - » Drone Shakti Scheme - Financial incentives to startups that are developing drone based solutions.
 - » Introduction of Drone Certification Scheme
 - » Initiatives to promote the use of drones in different sectors like Agriculture, infrastructure logistics etc.
- **Progress:** (ESI 2023-24)

- » India has established 109 training organizations and issued of 10,603 remote pilot certificates. Further, 22,943 unique identification numbers for registered drones, and 67 DGCA-approved Type Certificate for drone models.

- **Way Forward:**

- » **Create a favorable regulatory environment**: The 2021 rules need further simplification to promote ease of doing business in the drone sector
- » **Support Drone Industry**
 - **Launch** a drone startup accelerator program
 - **More R&D Support**: Government should incentivize/support R&D in drone sector to ensure innovation and development in the drone sector.
 - Organized drone-expos and conferences.
- » **Encourage PPP models in Drone sector** - This will help pool resources and expertise and accelerate the development of the sector.
- » **Human Resource Development**: Education and Skill development at graduate, post graduate and doctorate levels. Skilling opportunities related to drone sector repairs will be a big deal.
- » **Promote Industry-Academia collaboration** to support research.
- » **Focus on Cyber Security**: Ensuring that the drone system software is regularly updated.
- » **Plan for sustainable disposal of e-waste** generated by the use of drones.
- » **Encourage drone use in various sectors** like logistics, infrastructure, agriculture etc.

1) POWER SECTOR IN INDIA:

- **Introduction**
 - » Power is one of the most critical component of infrastructure crucial for economic growth and welfare of nations. India's power sector is one of the most diversified in the world. Sources of power generation ranges from Conventional Sources (Coal, Lignite, natural gas, oil, nuclear power) etc to non-Conventional Sources such as wind, solar, and agriculture and domestic waste.
- **Current Situation**
 - » **Power Transmission** in India is connected into one grid running on one frequency with the inter-regional capability of transferring 1,18,740 Megawatts (MW). It is emerging as one of the largest unified electricity grids in the world.
 - » India is the third largest producer and consumer of electricity in the world.
 - » **Installed Power capacity** of India is 444 GW (193 Renewable and 217 GW Coal based)
 - India aims to achieve the renewable energy target of 500 GW by 2030.
 - » The **peak electricity demand** increased by 13% in FY24 to 243 GW in FY24. Government is taking several measures to make electricity supply sustainable and thus the maximum rise in electricity generation was recorded in renewable energy sources for utilities.
 - » **Per Capita Power Consumption in India** is 1255 KWh which is below the global average of 3,250 KWh
- **Recent government initiatives in power sector:**
 - i. **Revamped Distribution Sector Scheme (RDSS):** It aims to reduce aggregate technical and commercial losses to 12-15% by 2025; reduce the average Cost of Supply and the average revenue realized gap to zero by FY25 and improve quality, reliability, and affordability of power supply to consumers through a financially sustainable and operationally efficient distribution sector.
 - ii. **SAMARTH (Sustainable Agrarian Mission on Use of Agri-Residue in Thermal Power Plant)**
 - **Progress:** Biomass co-firing in NCR thermal power plant has reached 1.68% and efforts are underway to take it upto 5%.
 - iii. **Infrastructure status** to energy storage systems, including grid-scale battery systems.
 - iv. **States allowed additional borrowing** in lieu of the power sector reform.
 - v. **Saubhagya Scheme** has contributed to 2.86 crore household becoming electrified since the launch of the scheme in 2017.
 - vi. **One Sun, One World, One Grid Initiative:** The initiative aims to interconnect regional grid across the world to transmit renewable energy seamlessly. Presently, discussions are going on with Saudi Arabia, UAE, Sri Lanka, Myanmar and Singapore.
 - vii. **Promoting Energy Efficiency:**
 - **UJALA (Unnat Jyoti by Affordable LEDs for All)**, launched in 2015, focuses on replacing conventional and inefficient variants with LED Bulbs, LED tube lights etc.
 - **Significance:** As per the Ministry of Power, it has led to estimated energy saving of 48.42 billion Kwh per year, with avoided peak demand of 9.7 GW

and reduction in GHG emission by 39.30 million tonnes of CO₂. It has also saved around 20,000 crore rupees in consumer electricity bill.

- **Street Lighting National Program (SLNP)**: Launched in 2015 to replace conventional streetlights with smart and energy efficient LED streetlights.

- **Key challenges of the power sector:**

- i. **Loss Making Discoms**: Corruption, inefficiencies, electricity theft etc are some reasons which have led to power distribution companies facing lots of financial challenges.
- ii. **Fuel Supply Challenges**: Power sector in India is primarily dependent on coal and thus is vulnerable to supply disruption. Further, domestic coal in India has high ash and sulphur content, making it highly polluting.
- iii. **Poor Efficiency** due to underutilization of power plants and AT&C losses.
- iv. **Insufficient infrastructure**: The installed capacity is not enough for economy growth at the rate of 7 to 8 %.
- v. **Climate Change challenges**: Fossil fuel is the largest source of energy for India but contributes to emission of CO₂.
- vi. **Electricity subsidy** for agriculture purpose leads to inefficient utilization.
- vii. **Renewable energy suffers** from issues like grid infrastructure constraints; energy storage limitations etc.

- **Way Forward:**

- i. **Work towards improving financial situation of DISCOMS:**
 - » Privatization or more penetration of private sector.
 - » Rationalizing tariffs for the domestic sector and there should be timely tariff reforms.
- ii. **Improve coal supply chain** -> to ensure adequate and timely availability.
- iii. **Increase focus on alternative fuels** like biofuels, municipal waste, agriculture waste etc.
- iv. **Enhance the production of renewable energy:** Take steps to surpass the target of capacity of 500 GW by 2030.
 - » Focus on issues hampering roof top solar; fast track smart grid and installation of smart meters.
 - » Invest in development of affordable and sustainable energy storage.
- v. **Demand side steps:** Focus on energy conservation and efficiency to deal with the shortage of power.

4. RENEWABLE ENERGY

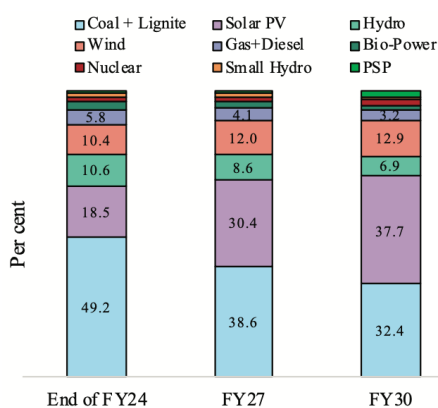
1) DATA: CAPACITY OF ELECTRICITY GENERATION (CURRENT AND FUTURE ESTIMATION)

- India is steadfast in its commitment towards non-fossil fuel-based energy resources with gradual transition from conventional to non-fossil fuel sources
 - » **Target:** In its revised updated NDC under Paris Agreement submitted to UNFCCC, India has set a target of achieving 50% cumulative installed capacity for generating electricity from non-fossil fuel-based energy resources by 2030.

As per the National Electricity Plan of the Central Electricity Authority, **non-fossil fuel (hydro, nuclear, solar, wind, biomass, small hydro, pump storage pumps) based capacity** which is **around 203.4 GW** (46% of the total) out of **441.9 GW** of total installed capacity in 2023-24 is likely to increased to **500.6 GW (64%)** in 2029-30.

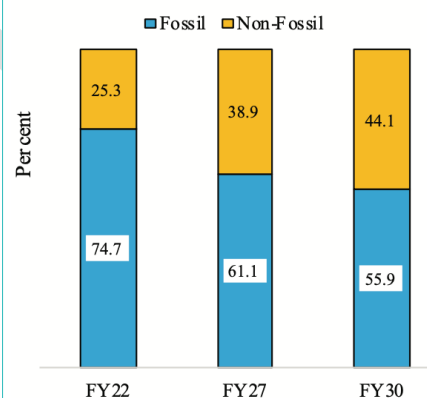
Note: As of 31st March 2024, non-fossil capacity is 203.4 GW which includes renewable capacity of 190.57 GW.

Chart XII.24: Percentage share of different sources in installed capacity of power



Source: Ministry of Power

Chart XII.25: Percentage share in power generation



2) RENEWABLE ENERGY: SIGNIFICANCE AND CHALLENGES

- Example Question:

- Discuss the significance of renewable energy in India's sustainable economic development. What are the factors hindering the growth of renewable energy sector in the country? Suggest some measures to deal with these problems. [15 marks, 250 words]

- Introduction

- Non-conventional energy sources refer to renewable energy sources which are replenishable at a rate faster than it is consumed i.e. they don't get depleted when used. These sources are also much less hazardous to environment compared to conventional sources of energy.
- **Popular Sources of Non-Conventional Energy Sources are:**
 - Solar Energy
 - Wind Energy
 - Tidal Energy
 - Geothermal Energy
 - Hydropower
 - Biomass Energy
 - Fuel Cell (Green Hydrogen)

Why do we need to invest more in the Renewable Energy Sector?

- **Energy Security and Reduced Import Dependency:** As for fossil fuels, India is mostly dependent on crude oil imports.
- **Economic growth:**
 - » Reduced cost of energy supply
 - » Reduced CAD -> Saving Forex
 - » Opportunity to emerge as a renewable energy hub for the world -> this can become an important source of employment and entrepreneurship.
 - » Improved export opportunity in the non-conventional energy sector.
- **Inclusive Development: Electricity in inaccessible areas** including where grid connectivity hasn't reached.
- **Environmental sustainability:**
 - » **Fighting Climate Change:** Meeting NDCs submitted at COP26 of UNFCCC.
 - » **Reducing Air Pollution and Water Pollution**
- **Strengthening India's Soft Power and Global Leadership:** India could also handhold other developing nations to explore the path of sustainable development via making best utilization of non-conventional sources of energy.

Steps taken by Government:

- **Increased Investment:** As per ESI 2023-24, the clean energy sector in India saw new investment of **Rs 8.5 lakh crores** between 2014 and 2023. The RE sector is expected to attract investments of about Rs 30.5 lakh crore in India between 2024-2030.
 - » **Steps to attract foreign investment:** FDI upto 100% under automatic route is allowed in renewable energy sector including offshore wind energy.
- **Ambitious Targets:** Revised NDC of India under Paris agreement targets 50% of electricity generation capacity by 2030 through non-fossil sources.
- **Promoting Ease of Doing Business:**
 - » **Waiving of Inter-State Transmission System (ISTS)** charges for inter-state sale of solar and wind power for projects to be commissioned by 30th June 2025.
 - » **Laying of new infrastructure:** New transmission lines and subs station capacity for evacuation of renewable power etc. under **Green Energy Corridor scheme** for evacuation of renewable energy
- **Sectoral Initiatives:**
 - » **Solar Energy:**
 - **Solar Parks Scheme:** Scheme has a sanctioned capacity of 39.7 GW for the development of 56 Solar Parks in 13 states.
 - **PM-KUSUM** - Focused on de dieselization of the farm sector.
 - **PLI Scheme** for National Program on High Efficiency Solar Photovoltaic (PV) Modules.
 - **PM Surya Ghar: Muft Bijli Yojna** - focused on installing rooftop solar in one crore households.
 - » **Steps to promote wind Energy:**
 - National Offshore Wind Energy Policy notified in 2015
 - National Wind Solar Hybrid Policy was adopted in 2018
 - » **Steps to promote bioenergy:**
 - National Policy on Biofuels, 2018

- The National Bio-energy program - notified in Nov 2022 - for setting up biomass power and cogeneration.
 - » **National Green Hydrogen Mission**: It aims to make India a green hydrogen hub, aiding to fulfill its target of production of **five million tonnes of green hydrogen by 2030** along with allied development of renewable energy capacity.
 - **International Collaboration**:
 - » International Solar Alliance
 - » Global Biofuel Alliance announced during 2023 G20 Summit in India.
 - » Global Renewables and Energy Efficiency Pledge of COP28 (UAE) - calls for tripling the rate of renewable capacity by 2030.
- **Problems faced by Renewable Energy Sector**
 - i. **Mobilization of necessary finance and investments on competitive terms**
 - ii. **Land Acquisition**: Solar and wind Power projects require large land areas, which has in recent years emerged amongst the biggest challenges faced by the sector.
 - For e.g. windmills and Solar plants are impacting the habitats of critically endangered the Great Indian Bustard.
 - iii. **Intermittent Nature of Renewable, Lack of energy storage, and wastage (15-20%)**
 - This makes grid integration difficult.
 - The absence of storage facilities leads to wastage to the tune of 15-20% of electricity generated.
 - iv. **Poor Manufacturing Ecosystem** in the country leads to import dependency for windmills and solar power panels.
 - v. **Lack of Skilled Manpower**: India still has not been able to attract youth towards this sector. Moreover, the opportunity to train in the sector is also negligible
 - vi. The **financial conditions of our distribution companies (discoms)** is another hindrance.
 - The additional solar and wind capacity will come from private sector and developers would be discouraged if they are not assured of market and timely payments.
 - vii. **International Trade Rules**: WTO trade rules have sometimes hindered our local manufacturing.
- **What further needs to be done**
 - i. **For increasing Finance**
 - More government investment
 - Creating mandates for provident funds to invest in infrastructure and environmentally sustainable projects
 - Increasing the priority sector lending limit for bank loans under solar energy from a meagre Rs 15 crore.
 - Mobilizing retail savings by way of tax exemption on the lines of Section 80CCF
 - Promote Green Bonds.
 - ii. Standardizing the definition of green to be able to target government efforts in the direction
 - iii. Promoting more R&D on Green sector
 - iv. Promoting local manufacturing
 - v. Dealing with intermittency and wastage (due to variability and lack of storage facilities)
 - Focus more research towards storage technology -> use tax incentives and VGF.
 - Promoting use of smart grid, net metering and storage devices.

- Redesign power markets to reflect the new feed for flexible supply and demand
- **Options available for storage:** Hydrogen based storage, lithium ion batteries, and pump storage plants.
- Hydrogen based storage is used for long term
- Lithium-ion batteries are suitable for day-to-day storage
- vi. **Dealing with Land Acquisition Issues:**
 - **Minimising total land-use requirements** for renewable energy by promoting offshore wind, rooftop solar and solar on water bodies
 - **Identification and assessment of land for renewable generation** by limiting undue regional concentration and developing environmental and social standards for rating potential sites.
 - **Attention on Indian agri-voltaics sector** — securing benefits to farmers and incentivising agri-voltaics uptake where crops, soils and conditions are suitable, and yields can be maintained or improved.
- vii. **Ensure Greater Synergy among all the concerned stakeholders** through a participatory approach and awareness generation, can further help to fasten the pace of shift towards renewable sources of energy.
- viii. **International Cooperation:**
 - Renewable Energy Technology should be declared as a global public good, including removing of IPR barriers to technology transfer.
 - Improve global access to supply chains for renewable energy technologies, components and raw material.

- **Conclusion1:** Time is not on our side and our climate is changing before our eyes. We need a complete transformation of the global energy system.
- **Conclusion2:** By addressing these challenges and implementing these measures, India can further accelerate its transition to renewable energy, promoting sustainable economic development while reducing its environmental footprint.

3) PROBLEM OF STORAGE IN RENEWABLE ENERGY

- **Introduction:**
 - » India is emerging as a renewable energy powerhouse with a target of 500 GW of installed non-fossil capacity by 2030. However, the country faces a major challenge in the form of lack of storage capacity.
- **Why is storage needed in renewable energy sector?**
 - » Handling Intermittency and variability problem of renewable due to sudden cloud cover and decreased wind velocity.
 - » Providing energy in non-solar/wind hours
 - » Improving grid stability
 - » Enabling larger renewable energy integration
- India's doesn't feel the pinch right now because we have more than 200 GW of coal based capacity which is managing the show by backing renewables in case of variability. So, decarbonization will be difficult without finding an alternative.

- **Various Storage options and their benefits and Limitations**
 - i. **Hydrogen based storage:** It is found feasible for long-term storage (cross seasons)
 - ii. **Lithium-ion batteries:** They are the ideal source for day-today storage.
 - **Concerns:** Though the cost of batteries have declined by 80% over the last decade, it is still quite expensive as the levelized cost of battery is about Rs 8 to Rs 10 per unit.
 - iii. **Pump Storage Plants:**
 - India has a total capacity of about 4.7 GW (out of a worldwide capacity of 149 GW). The CEA estimates that India's storage potential is of 100 GW.
 - **Concerns:**
 - High investment cost
 - Long Gestation Periods
 - Non-remunerative pricing models
 - Lack of adequate sites having the required topography
 - India's expertise in pump storage plants is somewhat a suspect with the Tehri pump storage project yet to be commissioned though construction began in 2011 and was to be complemented in 4.5 hours.
 - There may be environmental concerns associated with this kind of storage.
- In general, we can say that there are two main reasons for lack of storage capacities:
 - » **Cost** of storage technologies is still relatively high.
 - » **Regulatory Framework** for storage is not yet fully developed.
- **Some initiatives to deal with these challenges:**
 - i. **National Framework for Promoting Energy Storage System** - issued in Aug 2023 - for development and deployment of energy storage system to facilitate energy transition in the country.
 - ii. **Viability Gap Funding** for battery Energy Storage Systems.
 - iii. **Guidelines for the promotion of Pumped Storage Projects (PSPs)** was issued by Ministry of Power in April 2023.
 - iv. **National Storage Mission** launched in 2020 and wants to develop 100 GW of storage capacity by 2030.
 - v. **The Green Energy Corridor**: These are corridors which will be used to transmit electricity from renewable energy projects to load centres. These will also include storage facilities.
 - vi. The **National Battery Manufacturing Policy**, launched in 2020, aims to promote the manufacturing of batteries in India. This will help to reduce the cost of storage technologies.
- These are steps in the right direction, however, more efforts needs to be put to expand the capacities in parallel with the expansion of the renewable energy.
 - i. **More R&D in battery storage** to reduce the cost: Government should increase the financial incentive and subsidies to development and adopt more advanced energy storage technologies.
 - ii. **Grid Modernization**: To accommodate energy storage and intermittent renewable energy sources effectively. Implement smart grid technologies to enhance grid flexibility.

- iii. **Finalize a regulatory framework** which would help investors to develop and deploy storage projects.
 - iv. **Public Private Partnerships** - It can be used to develop and deploy storage projects.
 - v. **Demand Side measures** to reduce the need of storage e.g. smart metering, smart grid, energy efficiency etc.
 - vi. **Skill Development (Human Resource Creation)**: To develop workforce capable of designing, installing and maintaining energy storage systems.
 - vii. **International Collaboration** bilaterally and multilaterally to leverage best practices and sharing knowledge on energy storage solutions.
- **Conclusion**: These steps have the potential to increase the storage capacity for renewable energy sector and to help to make renewable energy a more reliable and affordable source of electricity.

4) DECENTRALIZED RENEWABLE ENERGY

- **Example Question:**
 - » "Decentralized renewable energy play a vital role in transitioning towards a more sustainable and resilient energy future, particularly in areas where centralized grid infrastructure is inadequate on non-existent" Critically Analyze [10 marks, 150 words]
- **Decentralized Renewable energy** refers to generation and distribution of renewable energy at a small and localized level, typically closer to the point of consumption. It can be generated via several renewable energy sources, including solar, wind, hydro and bioenergy.
 - » The DRE technologies include solar run textile manufacturing units, biomass powered cold solar storages and micro solar pumps etc.
- **Key characteristics** include localized generation, distributed energy resource, Off-Grid or Mini-Grid Solutions.
- **Advantages:**
 - » **Environmentally sustainable**: DRE doesn't lead to environmental damages which are caused by large scale hydropower plants, large scale solar parks etc.
 - » **Energy Security**: Technologies like rooftop solar, micro hydel power plants etc. can lead to sustainable energy production and thus can ensure long term energy security.
 - » **Inclusive Growth**: DRE can ensure energy supply and thus other associated services in remote, under-developed regions.
 - » **Scalability and Flexibility**: DRE can be scaled gradually depending on the local needs.
 - » **Job Creations**: DRE will stimulate local job opportunities in manufacturing sector, installation, operations etc.
 - As per MNRE, DRE has a market potential of Rs 4 lakh in rural and peri-urban communities in India.
- **Challenges and way forward**
 - » **Repairing** remained a challenge: Enough human resource and contact with manufacturers is still poor, especially in remote areas.
 - Skill development can not only improve the repair services but can also provide increased job opportunities.

- » **Increasing Affordability** - For a lot of people, affordability is a major concern and increased access to loans/government incentives is crucial for accessing the DREs.
 - At the same time, development of DRE industries in India, more R&D, etc. would be important for reducing the cost of the technology.
- » **Policy support** especially at state level: In India state level policy schemes are incapable of effectively supporting DRE technologies, especially when it comes to enabling affordable financing for entrepreneurs and end-users.
- » **International Cooperation** across sectors for an equitable exchange of knowledge and resources is essential for scaling DRE technologies.

- **Conclusion:**

- » DRE can play a vital role in transitioning towards a more sustainable and resilient energy future, particularly in areas where centralized grid infrastructure is inadequate or non-existent.

5) SOLAR ENERGY

- **Introduction:**

- » As per Ministry of Power, India's current (June 2024) Solar Energy Capacity is 85 GW. This means that India missed its ambitious target of 100 GW solar power generation capacity by 2022, but it also means that we need to accelerate our efforts to achieve the aim of reaching 185.6 GW of solar capacity by FY27 as per the National Electricity Plan.

- **Advantages of Solar Energy**

- » **Renewable Energy Advantages**

- (climate change, pollution, Energy Security, Economic Opportunities, Inclusive growth, cost competitiveness, reduce import dependency etc.)

- » **Solar's advantage over other renewable energy:**

- Available during office hours
- Longer life equipment
- Low running cost
- India's tropical climate is suitable for solar energy
- Less damaging than other renewables
 - A new study by WWF and IRENA found that wind and solar power are significantly less damaging than other renewable pathways.

- » **Solar Energy is becoming more and more competitive**

- **Government Efforts to promote Solar Energy**

- i. **Jawaharlal Nehru National Solar Mission (JNNSM)** - was aimed at achieving 100 GW of solar power by 2022 (60 GW through large and medium scale solar power projects and 40 GW of rooftop solar)
- ii. **Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects** - Conceived on the lines of "Charanka Solar Park" -> Act as flagship demonstration facility -> To provide power

developers with plug and play model, by facilitating necessary infrastructure along with all statutory clearance.

- **ESI 2023-24:** The scheme has sanctioned capacity of 39.7 GW for the development of **56 Solar Parks** in 13 states.
- **Update: June 2023:** The MNRE has extended the timeline for the 'Development of Solar Parks and Ultra Mega Solar Power Projects' Program to March 31st 2026 (FY26) from the earlier FY24.

iii. **Grid Connected Solar Rooftop Program**

iv. **PM Surya Ghar Muft Bijli Yojna**

- The finance minister, Nirmala Sitharaman, announced the launch of the 'Rooftop Solar Scheme' or the 'PM Surya Ghar Muft Bijli Yojana' in the 2024-25 budget.
- **Subsequently**, PM Modi launched the 'PM Surya Ghar Muft Bijli Yojna'.
- **Solar panels** are installed in the houses under the scheme to supply power to households and additional money for excess electricity output.
- The government will provide substantive subsidies directly to people's bank accounts and heavily concessional bank loans to ensure that there is no cost burden on the people.
- The scheme will help poor middle class to save upto Rs 15,000 to Rs 18,000 annually by getting free solar electricity and selling surplus power to electricity distribution companies.
- **Aim:** Under this initiative government aims to install solar power panels in one crore homes throughout the country.

v. **Pradhan Mantri Kisan Urja Suraksha Evan Uttham Mahabhiyan (PM Kusum)** aims at providing water and energy security to farmers and enhancing their income by making *Annadata* also a *Urjadata*. It focuses on creation of **3 things** - (A) Grid Connected Renewable solar power plants; (B) Installation of 20 lakh standalone Solar powered agri pumps; (C) Solarization of 10 lakh grid connected agri pumps.

- **Update:** In 2023, Government approved expansion of PM KUSUM with revised target of 49 lakhs pump to be installed/solarized under component B and C of the scheme.

vi. **CPSU Scheme (Phase-II) (Government Producer Scheme):** Aimed at setting up grid-connected solar PV power projects by PSUs and the Government organizations, using domestically manufactured solar PV cells and modules, with VGF support for self-use or use by the government or government entities.

vii. **New Solar Power Scheme (for PVTGs) Habitations/Villages:** Launched on 4th Jan 2024 under the **Pradhan Mantri Janjati Adivasi Nyaya Maha Abhiyan** for electrification of one lakh un-electrified PVTG households located in 18 states and one UT by provision of Off Grid solar systems where electricity supply through grid is not techno-economically feasible.

viii. **Steps to ensure easy finance for Solar energy**

- **Increased government spending.**
- **Financial Incentives:** Generation based incentives (GBIs), capital and interest subsidies, viability gap funding, concessional finance, fiscal incentives etc for providing financial support for various schemes have been initiated
- **New tax-free solar bonds.**
- Making roof top solar as part of housing loan by banks/NHB.

ix. **International Cooperation:** International Solar Alliance, OSOWOG etc.

- x. **New Innovations: Floating Solar Power Plant** (for e.g. the 100 MW plant at NTPC Ramagundam commissioned in July 2022) is being set up in various parts of the countries. It brings advantages like no land acquisition requirements, water conservation (less evaporation of water), less dust on power panel etc.
- xi. **Other steps to promote solar sector**
 - **Amendments to Electricity Act and Tariff Policy**
 - For strong enforcement for Renewable Purchase Obligations and for providing Renewable Generation Obligations (RGO).
 - Evacuation of renewable energy through **Green Energy Corridor project**.
 - Amendment to building by laws for mandatory provision of roof top solar for new construction or higher floor area ratio.
 - Infrastructure status to Solar projects.
 - Provision of rooftop solar and 10% renewable energy as mandatory under mission statements and guidelines for development of **smart cities**.
- xii. **Promoting Domestic Manufacturing of Solar PV cells and various components**
 - 1. **National Programme on High Efficiency Solar PV Modules**
 - It is a PLI scheme which aims to build an ecosystem for manufacturing of high efficiency solar PV modules in India, and thus reduce import dependence in the area of Renewable energy. It will strengthen Atmanirbhar Bharat initiative and generate employment.
 - Tranche-II of the PLI scheme was approved by cabinet in Sep 2022.
 - 2. **Other Older Steps:**
 - **Domestic Content Requirements:** Under schemes like PM KUSUM and Grid-connected rooftop solar, there are provisions of domestic content requirements.
 - 3. **Custom Duties:**
 - Government has announced imposition of Basic Customs Duty on import of Solar PV cells and modules with effect from 1st April 2022.
- xv. **Promoting Competition among States:** SARAL index by MNRE ranks states in terms of their efforts to incentivize rooftop solar.
- xvi. **Skill Development:** Focus on skill development of workforce: "Suryamitra Scheme" launched in May 2015. To create 50,000 trained personnel within a period of five years

- **Some Problems faced by Solar Energy Sector**

- i. General Problems faced by renewable energy sectors like finance, technology, human resource, land, Global Trade Rules etc.
- ii. **Substandard Equipment and Lack of Accountability:** For e.g. in Assam, under DDUGJY, Solar kit was distributed to ensure rural electrification, but when problems occurred either in the panel or the batteries, even within the contract period, the replacements haven't happened.
- iii. **Poor Performance of Rooftop Solar Power Projects**
 - A parliamentary panel (March 2023) has attributed low installation of solar rooftop and wind energy projects as key reasons for the shortfall in achieving India's renewable energy capacity target of 175 GW by 2022.
 - **Key Reasons**
 - a. **Residential Roof top policy** wasn't very effective. Government's policy was primarily focused on industrial rooftop.

- b. **Lack of cooperation from state electricity utilities and distribution companies** across India as it could hurt their finances.
- c. **India still doesn't have uniform policies around net metering**
- d. **Lack of awareness amongst public**

- For e.g. not many people are aware that with their home loans they can get loans for solar rooftops too.
- Further, people are not aware of the financial incentives and attractive return on investment that taking up rooftop solar power solutions can achieve.

iv. **Import Dependency:**

- Much of India's solar PV manufacturing uses imported components with parts mostly sourced from China. In the Budget announced in Feb 2022, government has announced a Basic Customs Duty of 40% on modules and 25% on solar cell imports from 1st April.

v. **Intermittency -> Lack of energy storage facilities.**

- **Some Limitations of expanding Renewable Energy:**

- **Land Acquisition -> Ecological and biodiversity losses:** Even dry regions support native vegetation such as grass, herbs and shrubs. They are diverse, ranging from woodland savannas, scrubland and grasslands to rocky outcrops, ravines and dunes. They have some protected species like black buck and Great Indian Bustard.
- **Loss of Ecological services** - like sequestering of more carbon than if trees were planted on them. Grazing based livelihoods also support millions in India.
- **Displacement:** In Kutch, communities displaced from their traditional grazing lands by renewable energy projects have been protesting these projects.
- **Emerging E-Waste Challenge:** As per a report by International Renewable Energy Agency (IREA), India could be generating more than 4 million tonnes of photovoltaic waste by 2050.

- **Way Forwards**

- **Increased focus on Rooftop Solar:** Development of positive retail led ecosystem to promote rooftop solar.
 - » **Improved policy of rooftop solar** - Increased focus on awareness generation.
 - » **Prioritize smart grid and net metering**
 - » **Incentivize discoms** by providing them with performance linked incentives.
 - » **Improving Tender Process** under DDUGJY: Tender process should be based on a '**Quality Cost Based System**', a well-accepted methodology for vendor selection globally which evaluates bidders based on technical and quality scores before looking at cost.
 - » **More focus on creating skilled workforce:** Government tenders for solar rooftops should have clause for training locals in maintenance by teaching them the required skill sets, and generating job opportunities for them.
- **Agri-Voltaics** - Deploying solar panel in a manner which allows agriculture below it.
 - » **Advantages:** Reduces evapo-transpiration and saves water; cooling effect of plants improves the efficiency of the solar power panels.
- **Ensure effective implementation of Social Impact Assessment and Environmental Impact Assessment** in large scale projects.
- **Deal with photovoltaic waste:**

- » Pan India sensitization drive about photovoltaic waste.
- » More R&D in development of better recycling mechanisms.
 - Other steps discussed under Renewable Energy Section above.

6) FLOATING SOLAR POWER PLANT

- Floating solar power plants, also known as floating photovoltaic (PV) systems or floating solar farms, are solar installations that are built on bodies such as lakes, reservoirs, ponds and even the sea.
 - » For e.g. the 100 MW floating solar plant fully operationalized at NTPC-Ramagundam (Telangana) is the largest floating solar power plant in India. It was built by BHEL through a EPC contract.
- **Advantages:**
 - » Resolves land acquisition related challenges.
 - » **More Efficient:**
 - Easy plain shadow free terrain and less dust accumulated by solar power panels.
 - **Ambient Temperature** also improves efficiency of the photovoltaic cells.
 - » **Water Conservation** (less evaporation)
- **Challenges:**
 - » **More Expensive** to install
 - » Requires **specialized engineering** to ensure stability and durability in water environments
 - » **Environment and Biodiversity** concern also arise from the blockage of sunlight.
 - » potential impacts on aquatic ecosystems, and maintenance
 - » **Maintenance** consideration for equipment in aquatic environments
- Thus, floating solar power plants offer an innovative approach to expanding solar power capacity while utilizing underutilized water.

7) WIND ENERGY

- Current Wind Energy Situation in India

- » India has made significant progress in the development of wind energy and has emerged as the fourth largest producer in the world with a total capacity **45.89 GW** as on 31st March 2024. But India did miss the 2022 target of achieving **60 GW capacity**.
- » Wind energy in India is led by indigenous wind power industry and strong project ecosystem, operation capabilities and a manufacturing base of 18 GW per annum.

- New Target: 140 GW by 2030

- » **India's Potential:** As per National Institute of Wind Energy (NIWE) in Chennai, India has a **much higher wind power potential**. At a hub height of 120 meters, the potential is of 602 GW of onshore and 100 GW of fixed and floating offshore. A very remarkable fact is that half of the potential is located in wastelands.

- Why did India miss its wind energy target?

- » **Inconsistent policy environments** also discourage investment in the sector
 - For e.g. In 2016, the government **decided to withdraw 50 percent of accelerated depreciation benefit** to industry from 2017.
- » **Legacy challenges** - infrastructure and transmission bottlenecks, financial conditions of discoms etc.
- » **Moving from feed in tariff system to e-reverse auction in 2017.**
 - Competitive bidding at low tariffs led to wind energy exploitation only remaining feasible in Gujarat and TN. In these two states plant Load Factor (PLF) is higher due to high wind speed.
 - The **overemphasis** on low tariff meant that tariff caps set for centrally sponsored auctioned projects are often too low to make the projects bankable or economically viable.
 - In 2022, MNRE has said that it would be doing away with the practice of reverse auction. This is expected to bring some relief to the wind energy sector.
- » **Cost Escalation:** Due to increased commodity prices
- » **Cutthroat Competition:** especially from cheaper photovoltaic power
- » **Reduced Policy Attention:** Government's policy focus has been on solar energy and wind energy has been reduced to playing a second fiddle, despite the fact that it was an early mover in the renewable energy sector.
- » **Land Acquisition** is emerging as another major issue.

- What Further needs to be done?

- Clear, well defined policy**
 - Perhaps the keenly awaited Renewable Energy Law that has been in the making for quite some time now.
 - This would bring policy clarity and remove uncertainties in the sector.
- Repowering** - i.e. upgrading the capacity of the existing wind turbines to produce more energy: It can increase productivity and spur socio-economic benefits.
- Work on solving legacy challenges:** Strengthening transmission infrastructure, DISCOMS and Changing State land policies to simplify land acquisition.
- Develop Offshore wind roadmaps:**
 - The offshore wind measurement campaign can yield LiDAR data to identify bankable offshore wind zones.

- Promote engagement among decision-makers at the federal, state and local levels of government, civil society organizations, and local stakeholder communities to align offshore wind development strategies and promote a collective understanding of offshore wind's socioeconomic benefits.
- v. **Creating Export Potential for wind energy equipment**
 - Develop a technology exchange program and align Indian manufacturing based with global wind supply chain to create export-oriented opportunities.
 - Including these components in FTAs
- vi. **Focus on modular and micro-wind turbines:** These can produce electricity at low wind speeds (5-6 kmph).
 - These micro-turbines could be fixed on terrace like a TV aerial; on street and highway lights; agricultural pumps; traffic signals etc.
 - It will reduce the land acquisition problem and would be suitable for regions which get continuous slow wind speed.

8) OFFSHORE WIND ENERGY

- **What is offshore wind energy?**
 - » It refers to deployment of wind farms inside the water bodies. They utilize the sea wind to generate electricity. These wind farms either use fixed foundation turbines or floating wind turbines.
 - **A fixed foundation turbine** is built on shallow water, whereas a **floating wind turbine** is built in deeper waters where its foundation is anchored in seabed. Floating wind farms are still in infancy.
- **Target:** MNRE has set a target of installing 30GW by 2030.
- **Advantages of offshore wind energy parks:**
 - » Renewable Energy
 - » No land acquisition and land scarcity issues
 - » Offshore wind turbines are more efficient compared to onshore ones (wind speed over water bodies is high and is consistent in direction)
- **Steps Taken**
 1. **A Revised Strategy for development of offshore wind energy projects** has been issued in Sep 2023.
 - It indicates a bidding trajectory for installation of 37 GW capacity of offshore wind energy.
 2. The **offshore wind energy lease rules, 2023:** To regulate the allocation of offshore wind sea blocks to developers have been notified.
 3. **National Offshore Wind Energy Policy, 2015:** Paves the way for offshore wind energy development activities, in waters, in or adjacent to the country, up to the seaward distance of 200 Nautical Miles (EEZ of the country) from the base line.
- **Challenges:**

1. **General challenges of wind energy** as discussed above.
2. **Technological and Engineering challenges:** High Sea breeze, corrosive saltwater and challenging seabed conditions require technological and engineering advancements.
3. **Local substructure manufacturers, installations vessels and trained workers** are lacking in India. Offshore wind turbines require stronger structures and foundations than onshore wind farms. This can cause higher infrastructure cost.
4. **Maintenance** of offshore wind farms are more costly due to problems like cyclones.
5. **Grid Infrastructure and Integration** will be challenges as the power generated at sea has to be transmitted to the onshore grid.
6. **Environmental concerns:** Environmentalists are worried that noise from offshore wind farms could impact fish habitats.

- **Way forward to promote offshore wind energy:**

- » **MNRE should set up a specific Renewable Purchase obligation (RPO) targets for each state just like it does for solar.**
 - Note: Currently, there are two types of RPOs - Solar and Non-Solar
- » **Feed in Tariffs:** Discoms can adopt feed-in tariff (FiT) regulations and make offshore wind power procurement mandatory.
- » **Skill Development** - Capacity building programs, vocational training, and educational initiatives should be implemented to develop a skilled workforce for offshore wind energy.
- » **Infrastructure Building:** Initiatives to promote domestic manufacturing and supply chain management.
- » **Environmental Impact Assessment and mitigation** to prevent environmental degradation and protect biodiversity.

9) TIDAL ENERGY

- Natural rise and fall of tides caused by the gravitational influence of interaction between Sun, Moon and Earth can be used to harness electricity with the help of generators. This energy is called tidal energy. Here, a turbine is propelled by the flow of water to produce electricity.
- **Significance of Tidal Energy:**
 - » **Renewable** and all associated advantages (Low pollution, global warming, achieving Paris agreement targets etc.)
 - » **High Energy density** - the rotation ability of water is higher than that of air and thus tides can produce much higher amount of energy.
 - » **Predictable and steady** - as tide cycle doesn't vary a lot
- **Limitation:**
 - » **High initial capital cost** (for e.g. India started building two tidal power projects in Gujarat and West Bengal in 2007 and 2011, with an installed capacity of 3.75 and 50 MW respectively. However, both the projects were put on hold because of steep rise in cost)
 - » **Engineering challenges** -> corrosion challenge and complex design requirements
 - » **Limited geographical areas are suitable** -> only those areas where tidal range is significant (five meters/ 16 feet)
 - » **Environmental Impact** -> tidal barrages may have negative impact on biodiversity of the region.
 - » **Distant source of energy** -> generally the place of energy production and place of energy utilization are far away.
- **So far**, India doesn't have any functioning tidal power plants.
 - » Though, tidal energy was not mentioned in the renewable energy targets of 2022, the MNRE has said that all renewable energy sources of energy will be eligible for the 2030 target.
- **Potential in India:**
 - » As per the Parliamentary Standing Committee's Report - Tidal Power Development in India - India's tidal energy capacity is **12.5 GW** and wave energy capacity is 41.3 GW.
- **Way Forward:**
 - » **Detailed assessment** of various types of ocean energy and explore the practically exploitable potential.
 - » India needs to initiate R&D for tidal energy and for this Gol should increase the budgetary allocation of MNRE.
 - » **Set up pilot projects** on tidal and wave energy - it can act as model for expansion.
 - » Focus on engineering, skill development and mapping of areas which will be suitable for tidal energy.
 - » **Strengthen Environmental Impact Assessment norms** - to ensure that tidal energy projects are not damaging environment.

- Example Questions

- i. Discuss the key advantages of Biogas. What are the key initiatives by government of India to promote the creation of biogas plants in the country. [10 marks, 150 words]
- ii. *'Promoting biofuels in India is of strategic importance'* Discuss in light of the National Biofuel Policy 2018. [15 marks, 250 words]
- iii. Discuss the 3 generations of biofuels and their advantages and disadvantages. [10 marks, 150 words]
- iv. "National Biofuel Policy is trying to balance India's food security & energy security with farmer's income security". Discuss. [12.5 marks, 200 words]

- Introduction

- » Biofuel is a fuel that is produced through contemporary biological processes, such as agriculture and anaerobic digestion, rather than fuel produced by geological processes such as those involved in the formation of fossil fuels.

- Generations of Biofuel (3 important)

- i. **1G Biofuel:** They are produced directly from food crops such as wheat, sugar, vegetable oil and **even animal fat** etc.
 - » **Advantages** includes known simple tech, cost competitiveness with fossil fuels etc.
 - » **Criticisms** include food vs fuel debate etc.
 - » They are also known as **conventional biofuels**. Most common first generation bio fuels include:
 - **Biodiesel:** Extraction of vegetable oils (both edible and non-edible), with or without esterification, from seeds of plants like soybean, rape (canola) and sunflower.
 - **Bioethanol:** Fermentation of simple sugar from sugar crops (sugarcane) or starch crops (corn, wheat etc).
 - It accounts for around 2/3rd of total biofuel production in the country.
 - **Bio-Gas:** Anaerobic fermentation of organic waste and crop residue as energy crops.
- ii. **2G Biofuels:** Produced from non-food organic crops such as wood, organic waste, food crop waste and specific biomass crops. It includes use of non-food-crops technologies like **jatropha**-based fuels.
 - The advantages include use of wasteland, less impact on food security.
 - There are some limitations including high capital cost, advanced conversion technologies etc.
 - One well known second-generation technology is Lignocellulosic processing which uses forest material.
- iii. **3G Biofuels:** The source is based on improvements in the production of biomass.
 - They are produced from micro-organisms like algae
 - Algae act as low cost, high-energy and entirely renewable feedstock. It has impressive diversity and higher yield. Advantages include the ease of generating the biomass anywhere where sunlight and carbon is present.
 - Third generation biofuel has the potential to be more sustainable and have a lower environmental impact than first and second generation biofuels.
 - Some limitations include still developing tech, high technology cost, some poisonous algae etc.

iv. 4th and 5th Generation Biofuels

A. Fourth Generation:

- » It takes the advantage of biotechnology to **engineer special crops** such as algae (sometimes called oilgae) for biomass production. The aim is to engineer microorganisms to produce biofuels more efficiently, with higher yields, and with lower environmental impacts.

A. 5th Generation Biofuels

- » They are known as **electrofuels**. They are produced from microbial synthesis using renewable energy sources. In this process, micro-organisms use electricity as an energy source to convert carbondioxide into liquid fuels, such as ethanol or butanol.

Note: 4th Generation and 5th Generation Biofuels are in the early stage of development.

- Targets:

- **National Biofuel Policy, 2018** (as amended in 2021): Country wide blending target of 20% ethanol by 2025 and 5% biodiesel by 2030

- Why growth of biofuel sector in India?

- The spurt in ethanol production in India is **almost entirely policy driven**. Initiatives like Ethanol Blending Program, National Biofuel Policy etc. have created conditions for the growth in the sector.

- Advantages of Biofuels

- Renewable and Energy Security:** Biofuels reduce dependency on imports and thus also reduces India's vulnerabilities to price fluctuations.
- Fighting Pollution:**
 - Ethanol blended petrol reduces emission of pollutants like carbon monoxide.
 - Biogas is a much better fuel than cow-dung cake. It can also reduce the problems of poor sanitation.
 - Further, biofuels helps in moving towards newer vehicle emission targets (BS-IV and BS-VI).
- Reduction in Greenhouse gases:**
 - A report by NITI Aayog quotes potential reduction of GHG emission at the point of use, i.e. from the tailpipe of the vehicle.
- Lesser impact on climate change**
 - Biogas also reduces the emission of GHGs such as methane and Carbondioxide.
- Can Promote sustainable agriculture:**
- Increase farmer's income**
- Sanitation:** Waste to biogas
- Social impacts:** Biofuels like biogas can reduce the drudgery of women involved in collecting fuel wood and thus plays a role in women empowerment.
- Reduces Import Dependency and saving foreign exchange**

- Some Criticisms:

- » **Land Use Change** to grow biofuel crops may have its own negative impact.

- For e.g. the constant increase in demand from sugarcane based ethanol from Brazil has meant extensive deforestation of rainforests resulting in GHG emissions from Brazilian ethanol use that was about 60% higher than petrol.
- » **Food Security Issues:** The argument that only surplus sugarcane and rice are diverted to fuel production may not hold in the longer term.
- » **Water Challenges** - Sugarcane is a water intensive crop.

- **Steps taken by government to support biofuels**

- i. **National Policy on Biofuels, 2018**
- ii. **Pradhan Mantri Ji-Van Yojana (PMJY)**
- iii. **Oil CPSEs are setting up 2G ethanol bio-refineries** in the country at Panipat (Haryana), Bathinda (Punjab), Numaligarh (Assam), Bargarh (Odisha) and one demonstration project at Panipat..
- iv. **EBP and associated steps**
- v. **National Bio-Energy Program (FY 2021-22 to 2025-26):**
 - It comprises of the following schemes:
 1. **Waste to Energy Programme** (*Program on Energy from Urban, Industrial, and Agricultural Wastes/ Residues*) to support setting up of large biogas, BioCNG, and Power Plants (excluding MSW to Power projects)
 2. **Biomass Programme** (*Scheme to support manufacturing of Briquettes & Pellets and Promotion of Biomass (non-bagasse) based congregation in Industries*) to support setting up of pellets and briquettes for use in power generation and non-bagasse based power generation projects.
 3. **Biogas Programme** to support setting up family and medium Biogas in rural areas.
- vi. **Advisory to carmakers to introduce flexible fuel engines in Vehicles:**
- vii. **International Collaboration:** For e.g. the Global Biofuel Alliance (GBA) is one of the top priorities under India's G20 presidency.
 - Brazil, India and the USA, as leading biofuel producers and consumers of the world have agreed to work together towards the development of this alliance along with other interested countries.
 - This alliance will work towards facilitating cooperation and intensifying the use of sustainable biofuels, including in the transportation sector.

- **Way forward:**

- » The proposed expansion of 1G biofuel need to think about broader land use strategies by developing clear and detailed criteria for identifying land suitable for energy crops.
 - Crucial factors like GHG emissions, local pollution, food security, land laws, and resource availability must also be considered.
- » **For 2G Biofuels -**
 - **Locating the industries near feedstock** availability: It is important to map feedstock location and then plan biofuel production capabilities.
 - **More R&D:** Scope for the schemes **like Pradhan Mantri JI-VAN** Yojana needs to be expanded, and international grants and loans should be redirected towards 2G fuels.
- » India also needs to **develop alternative feedstock** for biodiesel production.
 - Millions of tonnes used cooking oil can be channelized here.

- **Conclusion:**

» Thus, the expansion of the biofuel sector needs to be multipronged and rooted in sustainability.

11) ETHANOL BLENDING

- Process of mixing ethanol with Petrol is called Ethanol blending. The mixture is called as Ethanol Fuel/ Gasohol which is considered as a quasi-renewable energy.
- In India, the practice of blending ethanol was started in 2001. Ethanol blending was first time mentioned in the Auto Fuel Policy of 2003.
- **Benefits of Ethanol Blending**
 - » Reduces vehicular emissions especially CO (Carbon Monoxides) emissions.
 - » It is cheaper than petrol as it is easier to manufacture.
 - » It reduces our import dependency.
 - Trade balance, foreign exchange etc.
 - Ethanol has higher octane rating than ethanol-free petrol
 - » In case of India Ethanol production can generate higher sugarcane prices for farmers.
- **Ethanol Blended Petrol (EBP) Program** was launched by the Government in **2003**, and was aimed at promoting 5% blending of molasses-based ethanol with petrol, to promote the use of alternative and environment friendly fuels, to reduce import dependency for energy requirements and to increase value addition to Sugar industry enabling them to clear cane price arrears of farmers.
- **Target for Ethanol blending of Petrol was 10% by 2022 and 20% by 2030.**
- The **Roadmap for Ethanol Blending** was released by June 2021 by PM Modi.

12) NATIONAL POLICY ON BIOFUELS, 2018

- **Aims**
 - » Reduce India's oil import dependency.
 - » Provide better income opportunities to farmers by helping them dispose of their surplus stock in economic manner.
- **Key Highlights**
 - » **Aim (as amended in 2021):** Country wide blending target of 20% ethanol by 2025 and 5% biodiesel by 2030
 - » **Categorization** of Biofuels to enable extension of appropriate financial and fiscal incentives under each category.
 1. **Basic Biofuels** viz. First Generation (1G) bioethanol and biodiesel
 2. **Advanced Biofuels** - Second Generation (2G) Ethanol, Municipal Solid Waste to drop-in fuels, Third Generation (3G biofuels), bio-CNG etc.
 - » **Expands the scope of raw material for ethanol production**

- Allowing use of sugarcane juice, sugar containing materials like Sugar Beet, Sweet Sorghum, starch containing materials like Corn, Cassava, Damaged Food grains like wheat, broken rice, Rotten Potatoes unfit for human consumption for ethanol production.
- » **Allows use of surplus food grains for production of ethanol** for blending with petrol
- » **A VGF for 2G Ethanol bio-refineries** of Rs 5,000 crore in 6 years in addition to additional tax incentives, higher purchase price as compared to 1G biofuels.
- » The new policy encourages setting up of supply chain mechanisms for biodiesel production from non-edible oilseeds, used cooking oil, short gestation crops.
- » **Specifies the role of all the concerned ministries/ Departments** with respect to biofuels
- **2022 Amendment:**
 - » Advance the deadline to reach the blending target of 20% bioethanol in petrol, from 2030 to 2025-26.
 - It will promote the production of biofuels in the country, under the Make in India program, by units located in SEZs/Export Oriented Units (EOUs).
 - » Make additional feedstocks eligible for the production of biofuels.
- **Analysis: Expected Benefits**
 - » The new policy addresses **the supply side issues** that had discouraged the production of bio-fuels within the country.
- **Conclusion**
 - » Biofuels in India are of strategic importance as it augurs well with the ongoing initiatives of the Government such as Make in India, Swachh Bharat Abhiyan, Skill Development and offers great opportunities to integrate with ambitious target of doubling farmers' income, import reduction, employment generation, waste to energy creation.

13) NATIONAL BIO-ENERGY PROGRAM (NBP)

- Notified by MNRE in Nov 2022.
- It is to be implemented in two phases - From FY22 to FY26.
- **Phase-1** of the program has been approved with a budget outlay of Rs 858 crores.
- **The NBP** comprises of following sub schemes:
 - i. **Waste to Energy Program:** (*Programme on Energy from Urban, Industrial and Agricultural Wastes /Residues*) to support setting up of large Biogas, BioCNG and Power plants (excluding MSW to Power projects). IREDA is the implementing agency for the program.
 - ii. **Biomass Programme:** (*Scheme to Support Manufacturing of Briquettes & Pellets and Promotion of Biomass (non-bagasse) based cogeneration in Industries*) to support setting up of pellets and briquettes for use in power generation and non-bagasse based power generation project.
 - iii. **Biogas Programme:** To support setting up of family and medium size biogas in rural areas.
- **Progress:**
 - As of 31st March 2024, installed capacity of biomass power and cogeneration projects was about 9.4 GW (grid connected) and 0.92 GW (off grid), waste to energy projects capacity was 249.74 MW

(grid connected) and **336.06 MW** (off grid). Under biogas program, about 51.04 lakhs of small biogas plants and 349 medium size biogas plants (10.6 MW) have been installed.

14) BIOGAS

- The biogas plants can generate biogas from organic substances like cattle dung, night soil, and biomass from the Kitchen, garden, farms and other such bio-degradable material. This biogas is generated through a process called anaerobic digestion (AD).
- **Advantages of Biogas:**
 - » **Renewable and Sustainable** (plus other related advantages -> energy security, reduced import dependency etc.)
 - » **Reduction in Greenhouse emission:**
 - » **Waste Management:** Organic waste gets converted into energy and leads to environmental cleanliness.
 - » **Versatility:** Biogas can be used for cooking, heating, transportation etc.
 - » **Rural Development and Employment** - Energy source in rural areas leads to these advantages
- **Key challenges:**
 - » Level of waste segregation still leaves a lot to desire.
 - » Maintenance of biogas plants and other equipment have remained an issue.
- **Steps taken to Promote Biogas**
 1. **Gobar Dhan Yojana** (announced in Budget 2018-19) – It aims to collect waste and solid waste from farmers to sell to entrepreneurs, who thereafter produce manure, biogas, and bio-CNG from the waste. Thus, the scheme connect farmers to buyers.
 2. **Unified Registration Portal for GOBARdhan (June 2023)**
 - » Ministry of Jal Shakti has launched the Unified Registration Portal for GOBARdhan. It will act a one stop repository to access investment and participation in Biogas/CBG sector at Pan India level and more importantly streamline the process of setting up CBG/Biogas plants in India.
 3. **National Biogas and Manure Management Program (NBMMP)** aims to set up family type biogas plants.
 - » **By MNRE.**
 - » **It is a central sector scheme with an aim to setting up family type biogas plants for providing biogas as clean cooking fuel and a source of lighting primarily set up for rural and semi-urban households.**
 4. **Promotion of off-grid biogas power project for captive power generation**

15) BIO-CNG (ALSO KNOWN AS COMPRESSED BIO GAS)

- **BioCNG**, also known as biomethane, is a renewable and clean-burning transformation fuel produced by upgrading biogas to natural gas quality. It's essentially purified biogas, made from organic materials like: Agriculture waste, food waste, and Sewage Sludge.
- It is an upgraded version of biogas (the dung-based version of which serves as cooking fuel in many villages in India).

- **How Bio-CNG is produced?**

- » **Pre-Treatment:** The waste is passed through a filter to remove hard material like Coconut shells and pieces of wood.
- » **Shredding:** the waste is shredded in a hammer mill and made into a slurry with water.
- » **Hydrolysis:** The slurry is kept in the pre-digester tank in aerobic conditions for one-two days to attract microbes - the process is called **hydrolysis.**
- » **Methanogenesis:** After hydrolysis the slurry is transferred to anaerobic digester where it is retained for 20-25 days. It is during methanogenesis that biogas is generated. This gas contains 65% methane, while the rest is Hydrogen Sulphide, carbondioxide, and water vapour.
- » **Purification:** The above gas is passed through a wet and dry scrubber to remove hydrogen Sulphide and CO₂. Methane, purified upto 95% is obtained here which is then compressed at high pressure in cylinders and send to filling stations. This highly purified methane is similar in chemical properties to CNG derived from petroleum sources and can thus be used in vehicles.

- **Advantages:**

- » **Renewable:** The energy source is renewable and thus reduces India's import dependency while ensuring **Atmanirbharta and Make in India.**
- » **Swatch Bharat:** The production of Bio-CNG from biodegradable waste, agricultural residue, cow dung and chicken litter etc. can contribute to sanitation goals.
- » **Fighting Air Pollution:**
 - Bio-CNG helps deal with air pollution on three levels - curtail methane emission, ward-off waste burning and phase out fossil fuel powered vehicles.
 - For e.g. the CBG plant inaugurated in Sangrur, Punjab will soon be processing 300 tons per day of paddy straw and produce 33 tonnes per day of CBG.
- » **Strengthening Rural Economy, organic farming -> More income to farmers and More jobs in rural areas**
- » **Decentralized energy** as it is produced closest to the point of consumption.
- » **No Intermittency** like solar and wind as CBG could be produced at all hours.

- **Limitation:**

- » **Maintenance cost** of Bio-CNG based vehicles is higher.
- » Further, users have complained that calorific value of Bio-CNG is lower than CNG as it contains moisture.
- » **Biogas plants** are also sometimes seen as methane bombs as any emission of leak from digester or pipelines can contribute to climate change significantly.

- **Government Initiatives:**

- » **SATAT (Sustainable Alternative Towards Affordable Transport) Initiative:**
 - Launched in 2018, it aims to promote production and use of Bio-CNG (Compressed Bio-GAS) in India. Under it, government sets up compressed Bio Gas (CBG) production plants and make available CBG in the market for use in automotive sector.
- » **National Bio-Energy Program** (FY 2021-22 to 2025-26)
- » **Compressed Biogas (CBG) Blending Obligation (CBO)** - Government has announced a phased compulsory blending obligation of CBG in CNG and PNG segments of the City Gas Distribution (CGD) sector in Oct 2023.
 - **CBO** is voluntary till 2025 and mandatory obligation will start from FY26.

- It shall be 1% (FY26), 3% (FY27) and 4% (FY28) respectively.
 - It shall be 5% for FY29 onwards.
- » Asia's largest compressed bio gas plant was inaugurated in Sangrur by Union Minister Hardeep S. Puri.

- **Parliamentary standing committee has highlighted key challenges in the implementation of SATAT initiative:**

- » **Governance Issues** - Multiple regulatory approval needed from MoEF&CC, MNRE, MPNG hinders entrepreneurs.
- » **Discontinuation of central financial assistance** from April 2021 - made it difficult for new entrepreneurs
- » **Oil PSUs have not shown interest** in the implementation of CBG projects.
- » **Difficulty in getting feed stock** - the bio-mass supply chain is highly unregulated

- **Way Forward:**

- » **National Biofuel Coordination Committee** should meet regularly to sort implementation issues.
- » Instead of Capex, generation-based incentives should be introduced
- » **Single window system** for regulatory approvals
- » **Capacity building** for ULBs

16) HYDROGEN FUEL CELL (S&T TOPIC)

- **Fuel Cell**

- » A fuel cell is a device that **converts chemical potential energy (energy stored in the molecular bonds) into electrical energy.**
 - A Proton Exchange Membrane (PEM) cell uses hydrogen (H₂), and Oxygen (O₂) as fuel.
 - The products of the reaction are water, electricity and heat.

- **Key Elements of a fuel cell**

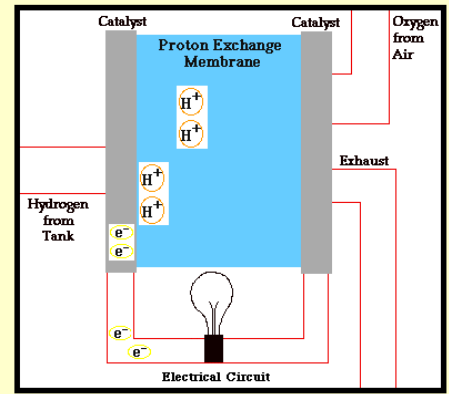
- » **The anode**, the negative electrode of the fuel cell, conducts the electrons that are fed from the hydrogen molecule so that they can be used in the external circuit.
- » **The Cathode**, the positive post of the fuel cell, has channels etched into that distribute the oxygen to all surface of the catalyst. It also conducts electrons back from the external circuit to the catalyst, where they can recombine with hydrogen ion and oxygen to form water.
- » The **electrolyte** is a **proton exchange membrane**. This specially treated material, which looks something like ordinary kitchen plastic wrap, only conducts positively charged ions. The **membrane blocks electrons.**
- » **The catalyst** is a special material that facilitates the reaction of oxygen & hydrogen. It is usually made up of platinum nano particles very thinly coated onto carbon paper or cloth.

- **How does a fuel cell work**

- » Hydrogen from a tank onboard the vehicle, enters into anode side of the fuel cell.
- » Oxygen pulled from air enters from cathode side

- **As the hydrogen molecule encounters the membrane, a catalyst forces it to split into electron and proton**

- » The proton moves through fuel cell stack and the electron follows an external circuit, delivering current to the electric motor and other vehicle components.
- » At cathode side, the proton and electron join again, and they combine with oxygen to form the vehicle's only tailpipe emission, water.



- Advantages of Hydrogen Fuel Cell

- » Fuel cell **avoids the "thermal bottleneck"** (a consequence of 2nd law of thermodynamics) and are thus inherently **more efficient than combustion engines**, which must first convert chemical potential energy into heat, and then mechanical energy.
- » Hydrogen is high in energy.
- » Fuel cells **don't have any moving part** and thus are more reliable than traditional engines.
- » **No pollution** (Only steam (H_2O) emitted as by product)
- » **No Greenhouse gas and Climate Change** (since no GHG are produced as bi-products)
- » **Ends dependency of Li-Ion batteries** (Please note for Lithium we are almost completely import dependent)

- Limitation

- » Complex and difficult to build
- » Still mostly in research phase
- » Extracting hydrogen is difficult and expensive - **catalyst used is Platinum**- which is very expensive.

17) HYDROGEN ENERGY

- **Hydrogen Energy** is a clean and efficient form of energy derived from Hydrogen (H_2). It has the potential to replace fossil fuels.
 - » Hydrogen can be produced from variety of sources including water, natural gas, and biomass.
 - » There are two main ways to produce Hydrogen:
 1. **Steam Reforming**: This process uses heat and steam to break down natural gas into hydrogen and carbon mono-oxide.
 2. **Electrolysis**: This process uses electricity to split water molecules into hydrogen and oxygen.
 - » **Cost of producing hydrogen** varies depending on the various methods used.
- It can be used in two primary ways:
 - » **Direct Burning** to produce heat and water
 - » **Fuel Cell Route** to directly produce electricity.
- **Advantages of Hydrogen Fuel**:
 - » **Abundance**: It is the most abundant element in the Universe.
 - » **Energy Density** -> High
 - » **Can contribute in achieving Net Zero by 2050**
 - » **No Pollution** (only releases water)
 - » **Leading options for storing energy from renewables**

- » **Advantages of Hydrogen Vehicles (Fuel cell Stack) over other Electric Vehicles (Lithium-ion batteries))**
 - A fuel cell electric vehicle can be refueled in just 5 minutes. EV takes 30-45 minutes for charging.
 - Energy storage per unit volume and weight is higher in fuel cells than other types of electric vehicles.
 - EV battery materials are controlled by a few larger players. Scaled up hydrogen fuel cell will bring countries on equal footing.
 - EV batteries (like Lithium ion batteries) have still not been found viable for heavy vehicles like trucks.
- **Some limitations of Hydrogen fuel**
 - » Hydrogen molecule is not available in abundance on earth and is found in combination with other elements.
 - » Hydrogen technology is "yet to be scaled up". Tesla CEO Elon Musk has called fuel cell technology "mind-bogglingly stupid".
 - » Lack of fueling station infrastructure: There are fewer than 500 operational hydrogen stations in the world today, mostly in Europe.
 - » Safety is a concern: Hydrogen is pressurized and stored in a cryogenic engine. Some companies like Toyota and Hyundai have said that safety and reliability of hydrogen fuel tanks is similar to that of standard CNG engines.
- **Note: Various types of Hydrogen**: The most common element in nature is not found freely. It exists only combined with other elements, and has to be extracted from naturally occurring compounds like water (which is a combination of two hydrogen atoms and one oxygen atom). This process is energy intensive.
 - » Grey Hydrogen: Hydrogen produced from fossil fuels. This constitutes a bulk of hydrogen produced today.
 - » Blue Hydrogen: Hydrogen generated from fossil fuels with carbon capture and storage options
 - » Green Hydrogen: Hydrogen generated entirely from renewable power sources. Here electricity generated from renewable energy is used to split water into hydrogen and oxygen.
 - For e.g. an IIT-Madras team generated hydrogen from seawater using solar energy. (June 2023)
- **National Green Hydrogen Mission**
 - » **Ministry**: MNRE
 - » With a vision to make India an energy independent nation, and to decarbonize critical sectors, the Government approved National Green Hydrogen Mission in Jan 2023, with an initial outlay of Rs 19744 crores upto 2029-30.
 - » The mission will facilitate demand creation, production, utilization, and export of Green Hydrogen and mobilization of Rs 8 lakh crores of investment by 2030.
 - » **Likely Outcomes by 2030**:
 1. Green Hydrogen Production Capacity of at least **5 MMT (Million Metric Tonne) per annum**.
 2. Reduction in fossil fuel imports by over Rs 1 lakh crores and creation of over 6 lakh jobs.
 3. Renewable Energy Capacity Addition of about 125 GW and abatement of nearly 50 MMT of annual GHG emissions.
 - **Interventions**:

1. Under the **Strategic Interventions of Green Hydrogen Transition (SIGHT) Program**, two distinct financial incentive mechanisms - targeting domestic manufacturing of electrolyzers and production of Green Hydrogen - will be provided under the mission.
2. **Regions capable of supporting large scale production and/or utilization of hydrogen** to be developed as **Green Hydrogen Hubs**.

▪ **Policy Framework:**

1. **Development of an enabling policy framework** to support establishment of Green Hydrogen Ecosystem.
2. **Robust Standards and Regulations Framework**
3. **Public Private Partnership framework** for R&D (**Strategic Hydrogen Innovation Partnership - SHIP**) will also be facilitated under the mission.
4. **Skill Development Program**

- **Several Pilot Projects** by PSUs like OIL, NTPC etc. have been initiated for the production of Green Hydrogen.

- **Government unveils Green Hydrogen Standards** (Aug 2023)

» It outlines the **emission threshold for production of hydrogen** that can be classified as 'green'.

▪ **Well-to-gate** emission of not more than 2 kg CO₂ for per Kg H₂.

- The well-to-gate emission include water treatment, electrolysis, gas purification, drying and compression of Hydrogen.
- The scope of the definition encompasses both electrolysis based and biomass-based hydrogen production methods.

» **Significance:**

▪ The definition of green hydrogen brings a lot of clarity to the mission of making India a global green hydrogen hub.

- **Way Forward: Following are the essential actions to ensure the launch of hydrogen economy in India**

» **Promote Demand**

- **Identify high demand sectors** like green ammonia, oil-refining, heavy duty transport etc. where initial demand can be catalysed via public incentives.
- **Pioneer voluntary purchase mechanisms** for green hydrogen embedded products such as green steel or green fertilizers similar to RE100 initiatives, where corporates like Infosys or google pledged to run completely on green energy.

» **Reduce cost of production of green hydrogen:**

- **Work towards reducing renewable energy tariffs** -> 70% of the cost of production of green hydrogen is the cost of renewable energy.
- **Scale:** India should strive to **incentivise the giga-scale production of green hydrogen components**, like electrolyzers, to take advantage of the global demand-supply gap and reduce the local green hydrogen prices.

» India should **identify hydrogen production clusters closer to the renewable parks** to utilise near-zero cost excess peak power which can be diverted to hydrogen plants

- » **Promote R&D:** Policymakers must facilitate investments in early-stage piloting and the research and development needed to advance the technology for use in India.
 - » **Focus on Domestic Manufacturing:**
 - Establishing an end to end electrolyser manufacturing facility will require more steps than just an PLI scheme. India needs to secure the supply of raw material that are needed for this technology.
 - » **In the initial phase, Blend Green Hydrogen with Grey Hydrogen** (as grey hydrogen is much cheaper to produce)
 - » **Plan for large scale refuelling network**
 - » **Start on pilot basis for gated infrastructure** like airports, ports, warehouses
- **Conclusion1**
- » Scaling up the technology and achieving critical mass remains a big challenge. More vehicles on the road and more supporting infrastructure can lower costs. **India's proposed mission** is seen a step in that direction.

18) ELECTRIC VEHICLES

- **Why we need to promote electric vehicles in India?**
 - i. **Energy Security** (India imports around 80% of its crude oil requirements), reducing import dependency; reducing CAD; saving forex etc.
 - ii. **Transition to renewable** is simplified if vehicles run on electricity instead of fossil fuels.
 - iii. Dealing with **air pollution and Noise Pollution** in big cities
 - iv. **Reducing Greenhouse gas emission -> Achieve Paris Targets on Climate Change**
 - v. **Competitive Domestic Manufacturing Ecosystem: Technology Development and Make in India**
 - » In automobile sector, India couldn't be a pioneer and had to depend on other countries to bring in the technology in the country and initiate the development of the sector.
 - » It's important that in electric vehicle sector, India develops its own indigenous industry which not only serves our domestic needs but is also able to earn us export revenues in future
 - » The development of electric vehicle sector will **provide employment opportunities through 'Make in India'** across a range of skillsets and would also grow **export opportunities**.
 - vi. **Making Transportation less expensive:** E.g. travelling a distance of 100 km in a conventional vehicle costs about INR 435, whereas in an EV, INR 97.
 - vii. **Power sector growth**
 - viii. **EVs can act as storage for Solar Energy**

Thus, the development of electric vehicle sector will benefit all citizens, will promote 'Ease of Living' and enhance the quality of life.

Schemes / Programs / Policies / Other steps to promote EV in India

1. Subsidy, Tax Incentives and PLI

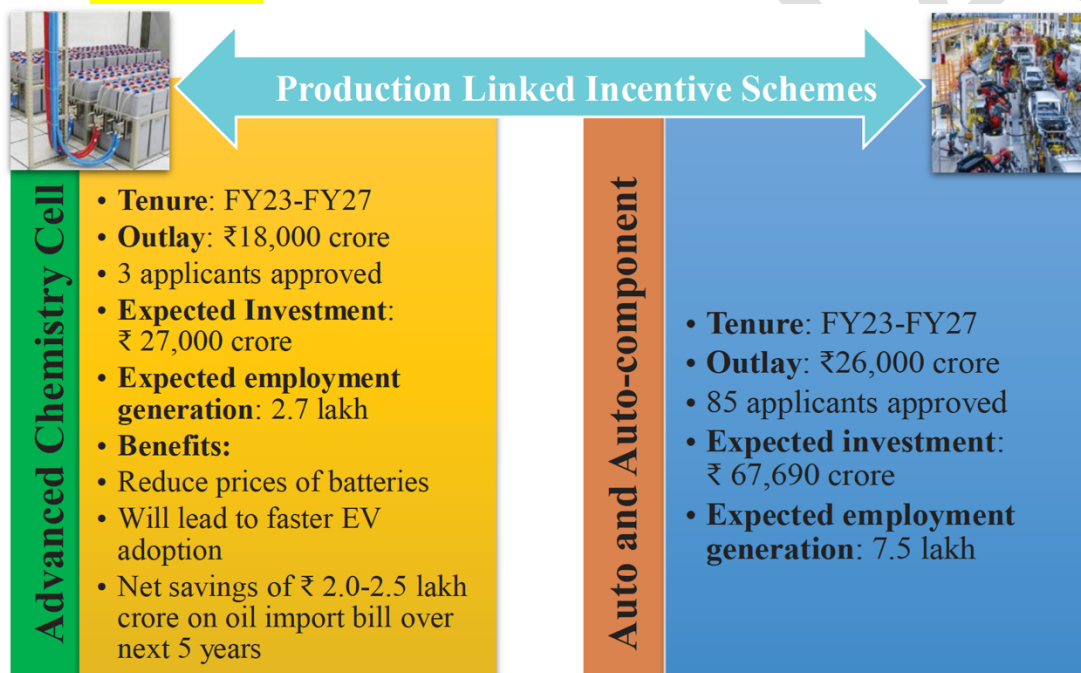
a. FAME (Faster Adoption and Manufacturing of Electric (& Hybrid) Vehicles):

- It is the key subsidy scheme for electric vehicles through which government incentivizes buyers to purchase e-vehicles.
- Phase-1 had started in 2015 and Phase-2 has been running since 2019.
- Under phase-2 companies may offer a discount of upto 40% on the cost of locally manufactured vehicles and claim it as a subsidy from government.

b. Tax Incentives:

- **GST on Electric Vehicles** is on the lower bracket of 5% as against 28% for conventional vehicles.
- Budget 2019-20: Tax Subsidies for EVs: Additional income tax deduction of Rs 1.5 Lakh on the interest paid on the loans taken to purchase electric vehicles before 31st March 2023.

c. PLI Schemes:



Source: Ministry of Heavy Industries

- In April 2023, government released SOP for the PLI scheme for the automobile and auto component industry.

2. **New EV Policy of 2024 to attract FDI in the sector** (see details in the next topic)
3. **National E-Mobility Program of Ministry of Power** (Launched in March 2018)
 - **Aggregate demand by procuring electric vehicles in Bulk to get economies of scale**
4. **National Mission on Transformative Mobility and Battery Storage** (approved by cabinet in March 2019)
 - The mission recommends and drive the strategies for transformative mobility and PMPs for EVs, EV components, and Batteries.

- The mission will ensure holistic and comprehensive growth of the battery manufacturing industry in India.

5. **E-Amrit Portal:** It is a web portal on electric vehicles and provide one stop solution for all information related to EVs - bursting myths around the adoption, details about policies/ subsidies etc.

- **Achievements of the above initiatives**

- » EV accounted for approx. 5% of total vehicle sales between Oct 2022 and Sep 2023.
 - In two wheeler and three wheeler section EV already has 50+ percentage of the new sales.
- » More than 200+ EV car models have been registered by the original equipment manufacturers (OEMs) availing benefits.
- » More than 6 lakh vehicles have been sold/supported under FAME.

- **Key challenges of India's EV Sector:**

1. **Battery Cost**

- Commercial EVs run on lithium-based batteries and India **imports most of Lithium and Lithium-ion batteries** making it very expensive.
- Therefore, India needs to **invest in research for battery technology and material sciences**.

2. **Low Lithium Reserves in India:** -> Chinese firms are already occupying stakes in countries such as Chile, Bolivia, and Australia which have substantive reserves, thus creating a kind of monopoly situation.

- Lithium reserves in India (Mandya Karnataka) will not be enough to keep up with the future demand.
- Further, exploration of lithium reserves would be **expensive**, and **hazardous to humans** as lithium is a rare earth metal found with other radioactive elements like beryllium, niobium, tantalum etc.
- Therefore, we need to **step up battery recycling capacity in India**. Battery Waste Management rules, 2022 has to be effectively implemented.
- Further, India needs to **invest in research into alternative battery technologies** using different metals.
 - **Aluminium, sodium, and zinc** based batteries have emerged as viable alternatives to lithium ones.
 - Aluminium based batteries could be beneficial for India as India is the fourth largest producer of aluminium and production has been consistently increasing. It is also cheaper than lithium, thus reducing the cost of EV batteries. Further, since Aluminium has higher valency (+3) than Lithium (+1), the aluminium batteries can have higher energy density.

3. **Limited availability of Charging Infrastructure and long time for EV Charging**

- Currently, India has about 2,000 charging stations operational and the number of EVs crossed 1 million in mid-2022.
- City planners, Municipal bodies, local administrative bodies, electricity companies, government etc. need to **build a comprehensive policy document** to meet future charging demands of EV.
- Further, the option of **battery swapping stations** also need to be **explored**.

- There are some practical challenges associated with this as the batteries need to be standardized and made interoperable with all EVs for easy removal and reattachment of battery back.
 - With improved batter swapping services and battery-as-a-service (BaaS) setting in, sales in the segment could see further growth.
 - **Provide information on public chargers to the users of EVs** through online maps and other means such as physical signage.
 - This will encourage increased use of adoption of EVs.
4. **Policy Conundrums:** Provisions of Electricity Act, 2003 is restrictive in nature and hinder setting up of charging stations - it is important to bring reforms here to promote ease of doing business.
 5. **Safety Issues:** The occurrence of some incidents involving EVs catching fire has raised doubts about its safety.
 - In response to this, MoRT&H have developed a new set of safety standards for EV batteries.
 6. **Other Infrastructure concerns:** Poor electricity distribution network; Surge in demand of electricity etc.
 7. **Human Resources:** The Indian auto industry is also facing a talent crunch as there aren't enough engineers with expertise in the field.
 8. **Subsidizing e-vehicles** is being criticized as some activists feel that it is subsidizing a small affluent section of the car-owning population, when there are better alternative to fossil fuels available.
 9. **E-vehicle alone will not decarbonize transport** -> we will have to decarbonize the electricity sector.
 10. **Public Awareness:** There is still very less understanding of EVs among the public, in terms of its benefits, risks, subsidies available, charging methods and tariffs, battery life, maintenance costs, and resale value.
 - A significant step to get rid of this problem is the - e-Amrit Portal.

- Conclusion

- » Electric vehicles have the potential to positively impact India's environment, climate target, economy and job market. We need to prepare ourselves by installing charging infrastructure, battery making factories and smart incentives for car companies and consumers to go electric.

19) GOVERNMENT'S NEW EV POLICY TO ATTRACT INVESTMENTS

- Why in news?

- » Government is considering extending the new EV Policy to include a retrospective effect (July 2024)
 - If this fructifies, it will extend benefits to entities who have already made their investments.

- Example Question:

- » What are the main objectives of the Electric Vehicle Policy announced by GoI in March 2024? Which EV companies will be eligible to get benefit under the new EV policy? [10 marks, 150 words]

- **In March 2024**, the GoI approved a new \$500 million EV Policy, offering range of incentives to Global EV Companies like TESLA, VinFAST etc. for investments.

- Main Objectives:

- » Attract foreign investment in EV Sector

- » Position India as a prime manufacturing hub for state-of-art EVs.
- » Providing Indian consumers with access to cutting edge EV Models, expanding the Make in India ecosystem, lowering cost of production, reducing oil imports, and reducing air pollution in urban areas
- » Fostering competitive domestic auto manufacturing industry: It will lead to sharper decline in costs of EV and faster adoption of EV.
- **Other expected advantages:**
 - » **Improve after sale services for EV sector**: Bain & Company's India EV Report (2023) observed that after-sale service is a **big pain point** for EV customers in India.
- **Eligibility Criteria and Benefits:**
 - » **Minimum Investment** of US\$ 500 million (INR 41.5 billion) withing the first three years of operation.
 - **Note (clarification by Gol)**: Timeline for investment are counted only after 240 days from the start of construction. (120 days application window and 120 days evaluation and confirmation by ministry)
 - If the US\$500 million is not completed within the specified three years, the company will not be eligible for the bank guarantee mentioned in the policy.
 - » **Achieving 25% Domestic Value Addition (DVA)** by the third year and 50% DVA by 5th year.
 - » **Customs duty of 15%** (as applicable to Completely Build Units (CBUs)) with a minimum cost, insurance and freight (CIF) value of US\$35,000 and above, for a five year period, contingent upon setting up manufacturing facilities within three years.
 - Earlier, it used to be 70-80%.
 - » **The duty forgone** on the total number of EV allowed for import would be limited to the investment made or INR 64.8 billion, whichever is lower.
 - » **Investment Commitment must be backed by a bank guarantee**, which will be invoked in case of non-compliance with DVA and minimum investment criteria.
- **Conclusion:**
 - » The policy is a recognition that import substitution for EVs, would require a layered and longer-sustained approach. At the same time to reap full benefit of FDI, the focus must be on ensuring that the critical components are made within the country, there is transfer of technology, and local manufacturing increases. Therefore, it is important to strengthen domestic players in the field too.

20) SODIUM ION BATTERIES (S&T TOPIC)

- **Practice Question:**
 - » Discuss the key advantages of Sodium ion battery technology for India [10 marks, 150 words]
 - » "Recent developments in developing sodium-ion battery technology might provide a solution to the materials crisis in the electric mobility transition" Discuss [15 marks, 250 words]
- **Introduction:**
 - Though, lithium-ion batteries are the most common type of batteries used in electric vehicles sector in India today, it has created a number of limitations for us.
- **Key Problems associated with Lithium-Ion Batteries:**
 - » **Resource Scarcity:**

- The most popular battery chemistry used by the global automotive industry currently is Lithium-Nickel, Manganese, and Cobalt (Li-NMC).
 - With increased demand for electric vehicles, Lithium has faced shortage and the prices per ton had touched US\$60,000 and above.
 - **Cobalt** mining (primarily from DRC) have led to persistent concerns regarding ethical mining.
 - **Nickel** prices have shot up in recent times due to Russia-Ukraine war.
- Due to these factors it is believed that there will be a shortage of batteries in 2024-25, followed by a lack of raw materials by 2027-28.
- » **Expensive:** Scarcity have led to these metals becoming expensive.
- » **Thermal Stability:** Lithium batteries have raised issues related to thermal stability and fire accidents.
- » **Chinese Domination and India's tense relation with China:**
 - China has amongst the world's largest proven reserve of Lithium, Chinese companies control mines in Australia and increasing in South America as well.
 - It also controls much of the Cobalt today in DRC.
 - Therefore, China also dominates the Lithium Cell Manufacturing with companies like CATL, BYD, and Gangfeng Lithium accounting for over three-quarters of global cell production. It also dominates the manufacturing of Charging system.
- Therefore, there is a gradual shift to Lithium Ferro Phosphate (LFP). This is because iron and phosphate are significantly cheaper and more environment friendly to extract and refine than the metals in L-NMC. However, it suffers from heavier weight and battery memory.
- **Sodium Ion Batteries and how it mitigates the above challenges for India:**
 - » **Easy Availability:** Na can be easily extracted from readily available salt and washing soda.
 - » **Less Corrosive:** Sodium is much less corrosive when compared to current version of Lithium being used. This will allow usage of cheaper materials like aluminium instead of copper both in the electrolyte as well as the battery frame.
 - The usage of these cheaper and lighter material will offset some of the weight disadvantages of Sodium over Lithium.
 - » **Doesn't use Cobalt:** Thus, the dependency on unethical mining concerns in Democratic Republic of Congo reduces.
 - » **Nickel** whose prices have sky rocketed is also used in much lesser quantity in Na-Ion batteries.
 - » **Thermal stability** of Na-Ion batteries is much higher than the Li-ion batteries reducing the chances of fire accidents which have occurred on many occasions in India.
 - » **Lower Cost** of lithium ion batteries may also lead to reduction in the cost of electric vehicles.
- **Limitations of Sodium-ion technology:**
 - Sodium is heavier -> Na-ion cell will be heavier
 - They are also less energy dense than li-ion cells. (For e.g. the latest generation of Li-NMC cells can store 250Wh/Kg as well as run at higher voltages which allows faster charging. Whereas the first generation Na-Ion cells is at just 160 Wh/kg currently and run at lower voltages.
 - Further, we are in the early stage of development of Li-ion batteries and many issues may come as more development happens. These may include number of cycles they can be charged before getting degraded; do they have the ability to be fast charged.

- **Way Forward:**
 - » **More R&D** to mitigate limitations of Sodium batteries.
 - » **Incentivize investments** in Na-ion technology by private entities with subsidies and PLI schemes.
 - » **Increased international collaboration on Lithium:** India-Australia can work together to mine Lithium deposits from western Australia.
- **Conclusion:**
 - » Both automotive and cell manufacturing industry are still learning, but it is imperative to not put all the eggs in the lithium basket specially India has no lithium resource.